Dhanush R. Kundur

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EDUCATION

The Ohio State University, College of Engineering — Columbus, OH
B.S. Mechanical Engineering | *Minor in Robotics & Autonomous Systems* — *Expected May 2029*
Dean's List: Fall 2025, Spring 2026 — GPA: 3.77 / 4.00

Relevant Coursework: Statics, Materials Science, Intro to ME Design (Arduino), MATLAB Programming, Engineering Economics, Linear Algebra & Differential Equations, Calculus I-III, Physics: Mechanics and E&M

PROJECTS

ATLAS: Autonomous Tracking and Landing Aerial System — Personal Project
Lead Designer & Developer — *May 2026 - Present*

- Designed, built, and flight-tested a 5-inch autonomous quadcopter end to end, owning the mechanical design, electrical assembly, and flight software on **ArduPilot**.
- Designed a 3D-printed **PETG** frame in **Onshape** with carbon-fiber arms and heat-set inserts (**Bambu Lab A1**); redesigned it as two pieces with a removable top plate after a crash.
- Soldered and configured the flight electronics (**H743** flight controller, 4-in-1 **ESC**, **GPS**/compass, **ELRS** receiver) and tuned **PID** gains for stable flight.
- Root-caused a takeoff flip by reading flight telemetry logs instead of swapping parts, tracing it to an inverted attitude-axis setting.
- Built a **Python**/**MAVLink** autonomy library (takeoff, waypoints, return-to-launch) and **OpenCV** target tracking, tested in **SITL**.

Dorm To-Do Board — ME 2900 Course Project
Individual Prototype — *Sep 2026 - Oct 2026*

- Built an **Arduino** task tracker integrating an LCD, rotary encoder, **I2C** real-time clock, servo flag, and RGB LED/buzzer alerts.
- Programmed a **C++** menu state machine with **EEPROM** saving so the list survives power loss and resets at midnight.
- Fixed LCD brownouts and an **I2C** lockup by isolating modules one at a time and splitting servo power from logic power.

All-Terrain Construction Cart — Ohio State University
Lead Designer & Prototyper | *4-person team* — *Jan 2026 - May 2026*

- Led design of a motor-assisted all-terrain cart on a 4-person team, scoped from 200K+ yearly construction lifting injuries.
- Down-selected a wagon concept via Pugh matrix; designed the frame, treaded wheels, brake, and assist motor in **Onshape**.
- Built and user-tested a working prototype, iterating the frame and load layout to reliably carry 20 lb loads over uneven terrain.

ENGINEERING & RESEARCH EXPERIENCE

Undergraduate Researcher, SOAR Lab — Columbus, OH
The Ohio State University | *Systems, Optimization, and Autonomous Robotics Lab* — *Aug 2026 - Present*

- Build and flight-test experimental UAV platforms for the lab's control and autonomy research under Dr. Debdipta Goswami (**Pixhawk Cube Orange**, **PX4**, **NVIDIA Jetson**).
- Own hardware-software integration on lab aircraft: airframe assembly, sensor and companion-compute mounting, power and signal wiring, and bench plus flight validation of each build before it is released for research testing.

Ohio State Baja SAE, Chassis Subteam — Columbus, OH
Chassis Subteam Member | *Single-seat off-road vehicle built to the SAE Baja rulebook* — *Aug 2026 - Present*

- Designing the chassis weld jig in **SolidWorks**, tolerancing locator seats so the welded roll cage matches CAD.
- Check tube sizing against SAE Baja rules and fabricate chassis parts in the team shop: tube notching, fixturing, and welding.

WORK EXPERIENCE

The Coder School — Ashburn, VA
Code Coach — *May 2024 - Aug 2025*

- Coached 10+ students weekly (ages 7 to 17) in **Python** and **Java**, teaching loops, functions, and OOP through custom lessons.

Assistant Researcher, MyEdMaster — Ashburn, VA
Medical NLP / Machine Learning — *Jan 2022 - Mar 2022*

- Co-developed a **Python** NLP pipeline on COVID-19 data (**Pandas**, **NLTK**, **spaCy**); co-authored a published paper (IJSSER, 2022).

SKILLS

CAD & Analysis: SolidWorks, Onshape, Fusion 360, ANSYS (FEA), GD&T fundamentals, engineering drawings
Manufacturing & Test: 3D printing (FDM, PETG/PLA), tube notching, fixturing, welding, soldering, bench and flight testing
Electronics & Embedded: Arduino, STM32, ArduPilot, PX4, Mission Planner, I2C, PWM, oscilloscope, multimeter
Programming: Python (NumPy, Pandas, OpenCV, pymavlink), C++ (Arduino), MATLAB, Java, Git/GitHub, Linux